Log 708 Mandatory 1 - Fall 2026

This is the first of two exercise sets with mandatory submission of solutions. The questions are to a large extent about using R in ways that we have seen in the introduction. We realize students are busy in the fall semester, and so have limited the amount of questions to be manageable in one good day. There is only one exercise, with several questions.

Submission date and work form.

You are free to work alone on the exercises, but we allow the cooperation of two students who can go together and submit a common solution. Mark solutions with both students' names/student numbers if you choose this. We want submissions in Canvas by Friday October 2. at 16.00.

Form of your report.

There are two preferred ways to prepare your report. In either case the first thing you should do is to set up a new R project, called "mandatory1" or something like that. Then submissions can be in *one* of the following forms.

1. Make an R script with the codes and comments necessary to solve the problems. Add a word file (or similar) with additional text where required. Submit both files.

OR:

2. Start using Rmarkdown. This requires *a little* additional initial preparation, but is **recommended**, in particular for students intending to do more data analysis beyond Log 708. Rmarkdown allows the integration in one document of R code, R output, ordinary text, and math formulas - with a very simple framework. All of the material for Log 708 has been prepared with Rmarkdown. This kind of integration allows a very productive workflow for anyone doing statistics or data analysis along with other academic work. At the end of this document is a description of what you need to do. Also a template Rmarkdown file will be published along with this exercise, so you can download the template, make a copy and start working on the copied file. You can produce output in html, word or pdf format. We suggest that you submit an html solution, but others are accepted as well.

Exercise 1

Some background

We are going to work on the data set `el_car_data_1.csv` (find the file in the "Data" module in Canvas). These data concern the "market shares" of various types of private cars in Norwegian municipalities. For 422 municipalities, (all except Svalbard), a number of variables were recorded in 2012 and then again in 2018. Among interesting things one can do is to compare the relative shares of car types between those two years. Norway has in this period had the highest growth of all countries when it comes to the share of electric cars. This is believed to have several different causes: There has been various powerful incentive regulations, where electric cars could pass for free on toll roads, pay half or even 0 ferry tolls, they could use lanes otherwise restricted to public transport vehicles. In addition the tax regulations have been favorable so as to make electric cars relatively much cheaper than conventional cars in Norway. (That's a reason why so many Tesla's are seen around here.) In addition, Norwegian consumers are relatively wealthy and many live in large houses with space for at least two cars, so an electric car is a great alternative as a nr. 2 car in a household.

In this exercise we are going to look at some aspects of the data, now mainly in a descriptive fashion. We may later use the same data material in deeper statistical analyses, to see if we can quantify some of the effects mentioned above. The idea would be that some municipalities are affected more by some of the incentives than others, e.g. public lanes is mainly relevant for municipalities close to major cities, ferry tolls are only relevant where there are ferries and so on. In the following we consider Oslo, Stavanger, Bergen, Trondheim to be "major cities". For neighboring municipalities to those, the option to use public lanes could be an important incentive to invest in an electric car.

In terms of data, the objects are pairs of (municipality, year). Among the variables which are not obvious from the name are:

- `id18` - a unique identifier for municipalities. This is needed because some names are used by more than one municipality!
- `income_med_adj` - the median income in a municipality, adjusted for inflation, i.e. incomes in NOK are adjusted to 2018 prices.
- `pct_expense` - the percentage of people who reported travel expenses for tax deduction in their annual tax declaration. Typically, such expenses are high for people who has to use ferry or pass toll road stations on a daily basis. So, in general, if `pct_expense` is high, it *indicates* that many people in the municipality have higher incentives to purchase an electric car.
- `sent_index` - a national index measuring the sentrality of municipalities. A description in Norwegian. Try google translate...
- `gasoline, diesel, el, hybrid, plugin` - the number of various car types
- `total` - the total number of cars

- `pct_elcar` the percentage of cars of pure electric type
- `zone` this is a categorical variable, meant to describe closeness to a major city. It has the following levels.
  - "city" if the municipality is among major cities
  - "below30"if a major city is within 30 minutes drive
  - "30to60" if a major city is between 30 and 60 minutes away
  - "above60" otherwise.

Start by downloading the file from the "Data files" module in Canvas, and save it with your other Log 708 data.

a. read the file into a data frame `cardata`. Check the first 10 rows of data, and the last 10. Names with "æ, ø, å" may look a little strange. Don't worry about this now. How many rows of data are there?

b. Calculate two new variables (columns) `pct_diesel, pct_gasoline` that are the percentages of diesel and gasoline cars in the municipalities.

c. It is best to have `zone` defined as an *R factor*. This means we give the vector "levels" corresponding to the possible values. Since `zone` is an *ordinal* variable, we can use the following code to arrange this.

```
cardata$zone = ordered(cardata$zone, levels = c("city",
                                                "below30",
                                                "30to60",
                                                 "above60"))
```

Run this code, check with `head(cardata$zone)` that you see some zones, and the levels.

d. make subsets `cardata12` and `cardata18` with data for the two years involved. Check that there are 422 rows of data in each.

e. In 2018 how many municipalities are in different zones? (Hint: `table(...)`). Make a barplot to visualize the distribution.

f. In 2018, what was the maximum and minimum population in municipalities, and what are the names of the two? (Can be solved with what we know, but you might want to check `?which.min`)

g. Think about: What do we measure when income (in NOK) is adjusted for inflation? Find the mean (over all municipalities) adjusted income in 2012 and 2018. (You can do this by considering each subset separately, or use `with(cardata, tapply(income_med_adj, year, mean)`

h. Now we are going to find the 10 municipalities with highest value for pct_elcar in 2012. This can be done with the following steps.

```
#find the indexes in pct_elcar in decreasing order wrt. pct_elcar
S <- order(cardata12$pct_elcar, decreasing = TRUE)
```

```r
#look at the first 10: '
S[1:10]

#extract the rows in cardata12 corresponding to the first 10 indexes.

cardata12[S[1:10], ]
```

i. Find the top 10 municipalities wrt. pct_elcar in 2018. Look at the `zone` variable for those 10. Does it show something related to the incentives mentioned above?

j. For 2018, let's see if we can find the mean value for `pct_elcar` by `zone`. Try to use `with` and `tapply` as we did for income by year above. The result should be striking.

k. The above differences can be visualized with boxplots. Try something like `with(cardata18, boxplot(pct_elcar ~ zone)`. In zone "above 60" there are a number of extreme observations. What could be characteristic for these municipalities? (This may require some knowledge of Norwegian road investment policy:-))

l. Try to find the total national percentage of elcars in 2012 and 2018. (You can not just take the average of `ptc_elcar` over all municipalities. (Why?). You need to find the total number of elcars and the total number of cars in Norway, and divide.)

m. For 2018, find correlation between `pct_expense` and `pct_elcar`. Also make a scatter-plot of the two variables, with reasonable text labels. Does it seem that the variable `pct_expense` to some extent can predict the amount of elcars in a municipality?

n. For 2018, find correlation between `sent_index` and `pct_elcar`. Also make a plot like above, and comment on this.

o. Repeat question n. for the variable `pct_diesel`. Does it look reasonable?

p. OPTIONAL: There are many more things you could do with these data, so feel free to make your own investigations. How is the `population` distributed among Norway's municipalities? Are diesel and gasoline cars replaced in the car population, or are el-cars just coming on top of conventional cars? ....

About R markdown.

As we have taken on the task of learning R, it will be highly rewarding to combine this with R markdown. As mentioned, this will allow us to combine R work with accompanying text remarks in order to create an integrated report. We can think of an R markdown file as an advanced R script with additional text processing options. As an R script allows you to save your code for later use, R markdown also can keep track of your ideas and thoughts while you were developing the code and working through part of a project. A suitably detailed introduction video is found in Canvas, along with the videos on basic R. On This link there is some basic info also.

To use Rmarkdown we need to do `install.packages("rmarkdown")`. An R markdown file is a text file with suffix `.Rmd` consisting of three types of input.

1. A short "yaml" header that sets some parameters for the document. Rstudio will provide this for us, and we can adjust it as needed.
2. Text in "markdown" style. This is ordinary text, but with some "tags" to make headings, lists, bold text, mathematical formulas etc.
3. R code chunks. These parts of the text file define R code, which at our choice can be shown in the final document or not, and which can be executed (or not) to show the output in the final document.

Once these things are done we can go to the Rstudio top menu, choose "Knit to html" for example, to produce an html document with our report. If you want to produce .pdf with math formulas you probably need to also run `install.packages("tinytex")` once. Then at the Rstudio console, write `tinytex::install_tinytex()`.

For preparing a report with solutions to this exercise set, a template document named `mandatory_1_template.Rmd` will be available with the exercises. You can use this as a starting point, and gradually delete parts and fill in your work. It is of course possible to work on an Rscript in parallell if you want to try out code there.

It can be handy to have a "cheatsheet" for Rmarkdown as well, you will find one at this page. It is a very compact presentation of almost all formatting commands in Rmarkdown.